

YOUR CAR SEES More Than You Think

Smart in-cabin intelligence is quietly reshaping the driving experience—detecting fatigue, fine-tuning comfort, and reinforcing safety without intruding on the journey.



YULIN WANG, Senior Technology Analyst at IOTechEx, has expertise in thermal management, mechanical modelling, signal processing, and software testing.



In modern vehicles, in-cabin sensing refers to the use of various sensors to monitor what is happening inside the car. These sensors focus primarily on two areas: driver monitoring systems (DMS) and occupant monitoring systems (OMS). DMS is used to track the driver's behaviour, such as attention, drowsiness, or engagement with the vehicle, while OMS monitors passengers and their positions. Together, these systems enhance safety and support new in-car features, making the driving experience both smarter and more responsive.

To build a complete picture of the cabin environment, modern vehicles rely on multiple types of sensors working in tandem. The most common are infrared cameras, which track faces and body movements. Some systems also use radars to detect presence and motion, torque

sensors to check if the driver's hands are on the wheel, and time-of-flight (ToF) sensors for tasks such as gesture control or depth sensing. Each sensor collects different types of data, which are then processed by dedicated algorithms. Together, they give the vehicle a clearer understanding of what is happening inside the cabin, enabling features like automatic alerts, personalised settings, or even emergency interventions.

As vehicles become increasingly intelligent, understanding cabin activity is no longer optional—it is essential. From tracking the driver's focus to recognising who is seated where, in-cabin sensing technologies are quietly transforming how cars respond to us. Often built on a network of discreet but powerful sensors, these systems offer not just greater safety but a more responsive and tailored driving